

FORENSICSHIELD COMPLIANCE REPORT

Report ID: RPT-20260913-5E9C2417
Case #: CAS-SIH-2026-001

1. Executive & Metadata Summary

Case Title:	Smart India Hackathon 2026 Demonstration Case	Generated At:	2026-09-13 08:56:23 UTC
Case Number:	CAS-SIH-2026-001	Generated By:	admin
Tool Version:	ForensicShield v1.0.0	Audit Chain Status:	INTACT

2. Forensic Status Classification Breakdown

Status Category	Verified	Inconclusive	Failed	Unsupported	Manual Review
Item Count	4	1	0	0	1

3. Read-Only Evidence Image Inventory

Evidence ID	Item #	Original Name	Size (Bytes)	SHA-256 Digest	Status
EVD-SIH-DISK-01	ITEM-001	synthetic_test_disk_	10,485,760	effcb0dbb243905b...	VERIFIED

4. Recovered & Carved Forensic Artifacts

Artifact ID	Type / Format	Source Offset	SHA-256 Digest	Confidence / Status
CARV-JPEG-001	Carved JPEG	10,240	hash_carv-jpeg-0...	Confidence: HIGH
CARV-PNG-001	Carved PNG	150,000	hash_carv-png-00...	Confidence: HIGH
CARV-PDF-001	Carved PDF	300,000	hash_carv-pdf-00...	Confidence: HIGH
CARV-ZIP-001	Carved ZIP	500,000	hash_carv-zip-00...	Confidence: HIGH
CARV-JPEG-TRUNC	Carved JPEG	800,000	hash_carv-jpeg-t...	Confidence: LOW

5. Technical Limitations & Forensic Scope

MANDATORY FORENSIC BOUNDARIES & TECHNICAL DISCLAIMERS:

- 1. **Storage Media Differences:** Magnetic HDDs support direct block overwriting. SSDs and NVMe drives utilize Flash Translation Layers (FTL), wear-leveling algorithms, and over-provisioned blocks. Logical sector overwriting does not guarantee sanitization of unmapped physical NAND blocks.
- 2. **Filesystem Boundaries:** Filesystem recovery is restricted to supported structures (FAT32, NTFS, Ext4). Damaged or unknown volumes require manual review.
- 3. **Signature Carving & Fragmentation:** Pure signature carving assumes contiguous block allocation. Carving non-contiguous fragmented files without allocation metadata produces corrupt byte streams.
- 4. **Validation Scoring Heuristics:** Validation scores (0-100) represent an empirical heuristic structural index and are NOT statistical probabilities of truth.
- 5. **Compliance Phrasing Notice:** Absolute claims ('100% unrecoverable') are excluded under ISO/IEC 27037 standards.